Informatica®

Informatica® Cloud Data Integration

Amazon DynamoDB Connector

Informatica Cloud Data Integration Amazon DynamoDB Connector
November 2024

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Preface

Use *Amazon DynamoDB Connector* to learn how to read from or write to Amazon DynamoDB by using Cloud Data Integration. You can also learn to create an Amazon DynamoDB connection, develop and run synchronization tasks, mappings, and mapping tasks in Cloud Data Integration.

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CHAPTER 1

Introduction to Amazon DynamoDB Connector

This chapter includes the following topics:

- Amazon DynamoDB Connector overview, 6
- Administration of Amazon DynamoDB Connector, 6
- Rules and guidelines for Amazon DynamoDB objects, 7

Amazon DynamoDB Connector overview

You can use Amazon DynamoDB Connector to connect Data Integration and Amazon DynamoDB tables. Amazon DynamoDB Connector enables you to read data from and write data to Amazon DynamoDB tables.

Amazon DynamoDB is a fully managed NoSQL database service that you can use to create database tables where you can store structured and unstructured data.

When you read data from or write data to Amazon DynamoDB tables, you can specify the Hosted Agent or Secure Agent. Create an Amazon DynamoDB connection to specify the location of Amazon DynamoDB sources, lookups, and targets you want to include in a task. You can use the Amazon DynamoDB connection in synchronization tasks, mappings, and mapping tasks. Create a synchronization task to synchronize data between a source and target. Create a mapping task to process data based on the data flow logic defined in a mapping or integration template.

Administration of Amazon DynamoDB Connector

As a user, you can use Amazon DynamoDB Connector after the organization administrator performs the following tasks:

- Enable the connector for the runtime environment. For more information about enabling connectors, see *Service and connector assignment* in the Administrator help.
- Provide user access to the Secure Agent directory that contains the success and error files. This directory path must be the same on each Secure Agent machine in the runtime environment.
- Create an Access Key ID and Secret Access Key on AWS. You can provide these key values when you create an Amazon DynamoDB connection.

Create an access key ID and secret access key

1. Log in to Amazon Web Services and navigate to the Security Credentials page.
2. Expand the **Access Keys** section, and click **Create New Access Key**.
3. Click the **Show Access Key** link.
4. Click **Download Key File** and save the file.

Rules and guidelines for Amazon DynamoDB objects

Consider the following rules and guidelines for Amazon DynamoDB objects when you configure source, target, and lookup objects in synchronization task, mapping, or mapping task:

- When you specify the create an Amazon DynamoDB target option in synchronization tasks, mappings, or mapping tasks, ensure that the source object contains only one primary key. While creating the table, the Secure Agent uses the primary key as the hash key in the Amazon DynamoDB table.
- Amazon DynamoDB Connector parses only the first 100 rows in an Amazon DynamoDB table. In the **Field Mapping** tab, you can view and map the fields available in the first 100 rows with the source and target objects.
- You cannot change the precision or data type for fields in the Amazon DynamoDB tables.
- You can change the scale of numeric data type fields in Amazon DynamoDB tables.
- You cannot specify multi-valued attributes and document data types as Amazon DynamoDB objects.
- When there is more than one data partition in an Amazon DynamoDB table, the Secure Agent can read the tables in parallel.

CHAPTER 2

Amazon DynamoDB connections

This chapter includes the following topics:

- Amazon DynamoDB connections overview, 8
- Amazon DynamoDB connection properties, 8

Amazon DynamoDB connections overview

Amazon DynamoDB connections enable you to read data from or write data to Amazon DynamoDB tables. You can use Amazon DynamoDB connections to specify sources and targets in synchronization tasks, mappings, and mapping tasks.

Create a connection and associate it with a synchronization task, mapping, or mapping task. Define the source and target properties to read data from or write data to Amazon DynamoDB tables.

You create an Amazon DynamoDB connection on the **Connections** page. Use the connection in the Mapping Designer when you create a mapping or in the Synchronization Task wizard when you create a task. You can specify the Amazon DynamoDB connection as source, target, or lookup in the synchronization task and mapping task.

Amazon DynamoDB connection properties

When you set up an Amazon DynamoDB connection, you must configure the connection properties.

Important: Amazon DynamoDB Connector is deprecated and has been moved to maintenance mode. Informatica intends to drop support in a future release.

The following table describes the Amazon DynamoDB connection properties:

Connection property	Description
Access Key	The access key ID used to access the Amazon account resources. **Note:** Ensure that you have valid AWS credentials before you create a connection.
Secret Key	The secret access key used to access the Amazon account resources. This value is associated with the access key and uniquely identifies the account.
Region	The AWS region associated with the account.

CHAPTER 3

Synchronization tasks with Amazon DynamoDB Connector

This chapter includes the following topics:

- Amazon DynamoDB sources in synchronization task, 10
- Amazon DynamoDB targets in synchronization task, 11
- Amazon DynamoDB lookups in synchronization task, 12
- Synchronization task example, 13

Amazon DynamoDB sources in synchronization task

When you configure a synchronization task to use an Amazon DynamoDB source, you can configure the source properties.

The source properties appear on the **Source** page of the Synchronization Task wizard when you specify an Amazon DynamoDB connection.

The following table describes the Amazon DynamoDB source properties:

Source Property	Description
Connection Type	Name of the source connection.
Source Type	Source type. Select **Single** for a single source object.
Source Object	Select the source object for the task.

You can also configure advanced source properties when you schedule the synchronization task.

The following table describes the Amazon DynamoDB advanced source properties:

Advanced Source Property	Description
Row Count	The maximum number of rows to read from the Amazon DynamoDB table.
Read Capacity Unit	The percentage increment of the read capacity unit. For Amazon DynamoDB Connector, a unit of read capacity represents two eventually consistent reads per second for items as large as 4 KB.

Read capacity unit

When you create or update a table, you can configure the provisioned throughput capacity using the read capacity unit for read operations. As your application data and access requirements change, you can increase or decrease your provisioned throughput by updating the read capacity unit.

You can use eventually consistent reads and not strongly consistent reads. For Amazon DynamoDB Connector, the read capacity unit represents the number of reads per second of items up to 8 KB in size per second. For example, if you specify 10 read capacity units, you are requesting a throughput of 10 eventually consistent reads per second of 8 KB for that table.

To modify the read capacity unit, specify a percentage increment of the read capacity unit when you configure the advanced source properties. For example, if the existing read capacity unit is 100, you can read 100 *8 KB of data per second from a table. To increase the number of reads by 20%, specify a value of 20 in the Read Capacity Unit field. You can now read 120 * 8 KB of data per second, which is 20% of existing unit of read capacity plus existing unit of read capacity.

Amazon DynamoDB targets in synchronization task

When you configure a synchronization task to write to an Amazon DynamoDB target, you can configure the target properties.

The target properties appear on the **Target** page of the Synchronization Task wizard when you specify an Amazon DynamoDB connection.

The following table describes the Amazon DynamoDB target properties:

Source Property	Description
Connection Type	Name of the target connection.
Target Object	Specify the target object for the task.
Create Target	Creates a target. Enter a name for the target object and select the source fields that you want to use. Default name is the source object name and by default, all source fields are used. The target name can contain alphanumeric characters and the period and the underscore special characters. It cannot begin with a numeric character. **Note:** You can create a target table only when the source table contains a single primary key.

You can also configure advanced target properties when you schedule the synchronization task.

The following table describes the Amazon DynamoDB advanced target properties:

Advanced Target Property	Description
Write Capacity Unit	The percentage increment of the write capacity unit. A unit of write capacity represents one write per second for items as large as 1 KB. **Note:** When you create a table during the synchronization task, the default unit of write capacity is 1.
Success File Directory	Directory for the success file. Specify a directory path that is available on each Secure Agent machine in the runtime environment. By default, Data Integration writes the success file to the following directory: `<Secure Agent installation directory>/apps/Data_Integration_Server/data/success`
Error File Directory	Directory for the error file. Specify a directory path that is available on each Secure Agent machine in the runtime environment. By default, Data Integration writes the error file to the following directory: `<Secure Agent installation directory>/apps/Data_Integration_Server/data/error`

Write capacity unit

When you create or update a table, you can specify the provisioned throughput capacity using the write capacity unit for write operations. As your application data and access requirements change, you can increase or decrease your provisioned throughput by updating the write capacity unit.

A unit of write capacity represents one write per second for items as large as 1 KB. For example, if you specify 10 write capacity units, you are requesting a throughput of 10 writes per second of 1 KB size per second for that table.

To modify the write capacity unit, specify a percentage increment of the write capacity unit when you configure the advanced target properties. For example, if the existing write capacity unit is 100, you can write 1 KB of data per second to a table. To increase the number of writes by 20%, specify a value of 20 in the Write Capacity Unit field. You can now write 120 KB of data per second, which is 20% of existing unit of write capacity unit plus existing unit of write capacity.

Amazon DynamoDB lookups in synchronization task

When you configure field mappings in a synchronization task, you can create a lookup to an Amazon DynamoDB object.

When you use an Amazon DynamoDB object as a lookup, you do not need to configure specific Amazon DynamoDB properties.

Synchronization task example

You work in a marketing firm and business analysts in your organization need to analyze trends based on tweets on Twitter.

You use Amazon DynamoDB Connector to consolidate the tweets from various users and store the unstructured data in Amazon DynamoDB tables.

Configure a synchronization task to consolidate the tweets based on username and store them in an existing Amazon DynamoDB table.

You perform the following synchronization task:

Define the synchronization task.

Configure a synchronization task to use the upsert operation.

Create a Twitter source object.

The source for the mapping is a Twitter object that contains the tweet details. The Twitter object is a single source in the synchronization task. The Twitter object includes the *TweetId*, *UserName*, and *Tweet* source columns. Specify *Tweets* as the resource for the source object.

Create an Amazon DynamoDB target object.

The target for the mapping is an Amazon DynamoDB table. The Amazon DynamoDB object includes the *UserName* and *AllTweets* target columns. Provide a name *test_twitter* for the target object and specify the connection type as Amazon DynamoDB.

Specify a data filter for the Twitter source object.

Define a data filter for the Tweets source object. For example, you can filter by `NoneOfTheseWords Equals 'Watch'`.

Configure a field mapping.

Map the fields under *Tweets* source data to the fields in the *test_twitter* target object.

To consolidate the tweets based on user name, enter the following lookup expression:
`concat((lookup( UserName=username ):test_twitter.tweets), Tweet`

When you run the task, the synchronization application writes the mapped source data to the target Amazon DynamoDB table.

Schedule the task.

You can schedule the task for each requirement and save. Configure the advanced source and target properties on the **Schedule** tab and save.

You can select the synchronization task from the **Explore** page and run the task. In **Monitor**, you can monitor the status of the logs after you run the task.

CHAPTER 4

Mapping and mapping task with Amazon DynamoDB

This chapter includes the following topics:

- Amazon DynamoDB sources in mapping task, 14
- Amazon DynamoDB targets in mapping tasks, 15
- Amazon DynamoDB lookups in mappings, 16

Amazon DynamoDB sources in mapping task

To read data from an Amazon DynamoDB table, configure an Amazon DynamoDB object as the Source transformation in a mapping task.

Specify the name and description of the Amazon DynamoDB source. Configure the source and advanced properties for the source object.

The following table describes the source properties that you can configure in a Source transformation:

Property	Description
Connection Name	Name of the source connection.
Source Type	Source type. Select one of the following types: - Single Object. - Parameter. Select **Parameter** to define the source type when you configure the mapping task.
Source Object	Source object for the mapping.

The following table describes the advanced source properties that you can configure in a Source transformation:

Advanced Property	Description
Row Count	The maximum number of rows to retrieve from an Amazon DynamoDB table.
Read Capacity Unit	The percentage increment of the read capacity unit.
Tracing Level	Sets the amount of detail that appears in the log file. You can choose terse, normal, verbose initialization or verbose data. Default is normal.

Amazon DynamoDB targets in mapping tasks

To write data to an Amazon DynamoDB table, configure an Amazon DynamoDB object as the Target transformation in a mapping task.

Specify the name and description of the Amazon DynamoDB target. Configure the target and advanced properties for the target object.

The following table describes the target properties that you can configure in a Target transformation:

Property	Description
Connection	Name of the target connection.
Target Type	Target type. Select one of the following target types: - Single Object. - Parameter. Select **Parameter** to define the target type when you configure the task.
Object	Name of the target object.
Operation	Target operation. Select one of the following operations: - Insert. - Update. - Upsert. - Delete. - Data Driven. Configure an expression to specify an update strategy based on the data. You can use one of the following constants in an expression: - DD_INSERT - DD_UPDATE - DD_REJECT - DD_DELETE **Note:** If you use the Data Driven operation, you cannot upsert.

The following table describes the advanced target properties that you can configure in a Target transformation:

Property	Description
Write Capacity Unit	The percentage increment of the write capacity unit.
Success File Directory	Directory for the success file. Specify a directory path that is available on each Secure Agent machine in the runtime environment. By default, Data Integration writes the success file to the following directory: `<Secure Agent installation directory>/apps/Data_Integration_Server/data/success`
Error File Directory	Directory for the error file. Specify a directory path that is available on each Secure Agent machine in the runtime environment. By default, Data Integration writes the error file to the following directory: `<Secure Agent installation directory>/apps/Data_Integration_Server/data/error`
Forward Rejected Rows	Determines whether the transformation passes rejected rows to the next transformation or drops rejected rows. By default, the mapping application forwards rejected rows to the next transformation.

Amazon DynamoDB lookups in mappings

In a mapping, you can configure a Lookup transformation to represent an Amazon DynamoDB object.

When you use an Amazon DynamoDB object as a lookup, you do not need to configure specific Amazon DynamoDB properties.

APPENDIX A

Amazon DynamoDB data type reference

This appendix includes the following topics:

- Amazon DynamoDB data type reference overview, 17
- Amazon DynamoDB and transformation data types, 18

Amazon DynamoDB data type reference overview

Data Integration uses the following data types in synchronization tasks, mappings, and mapping tasks with Amazon DynamoDB:

Amazon DynamoDB native data types

Amazon DynamoDB data types appear in the **Fields** tab for Source and Target transformations when you choose to edit metadata for the fields.

Transformation data types

Set of data types that appear in the remaining transformations. They are internal data types based on ANSI SQL-92 generic data types, which Data Integration uses to move data across platforms. Transformation data types appear in all remaining transformations in a synchronization task, mapping, or mapping task.

When Data Integration reads source data, it converts the native data types to the comparable transformation data types before transforming the data. When Data Integration writes to a target, it converts the transformation data types to the comparable native data types.

Amazon DynamoDB and transformation data types

The following table lists the Amazon DynamoDB data types that Data Integration supports and the corresponding transformation data types:

Amazon DynamoDB Native Data Type	Transformation Data Type	Description
Number	Decimal	Precision 1 to 28, scale 0 to 28
String	Varchar	1 to 104,857,600 characters
Boolean	Small Integer	Precision 5, scale 0
Binary	Binary	1 to 104,857,600 bytes

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